Data Analytics and Visualisation
CI7330 Coursework- Assignment 2

Name: Ayesha Shahzad
KU ID: K2446625
Date: 12/01/2026

Kingston University London

Faculty of Engineering, Computing and the Environment

Wine Quality Analysis and Hip Fracture Prediction

For both datasets, a comprehensive regression analysis has been conducted on RStudio and here are the key findings:

1. Red Wine Quality Analysis

1.1 Initial Exploration of the Dataset

The red wine dataset contains 1,599 samples with 11 physicochemical predictors and expert quality ratings (mean quality: 6.13, range: 3.1-8.9), indicating most wines are rated moderately. The distribution plot (figure 1) shows a strong peak at 6, with more wines rated 7 than 5, suggesting the sample skew slightly towards above-average quality. The dataset is complete, with no missing values.

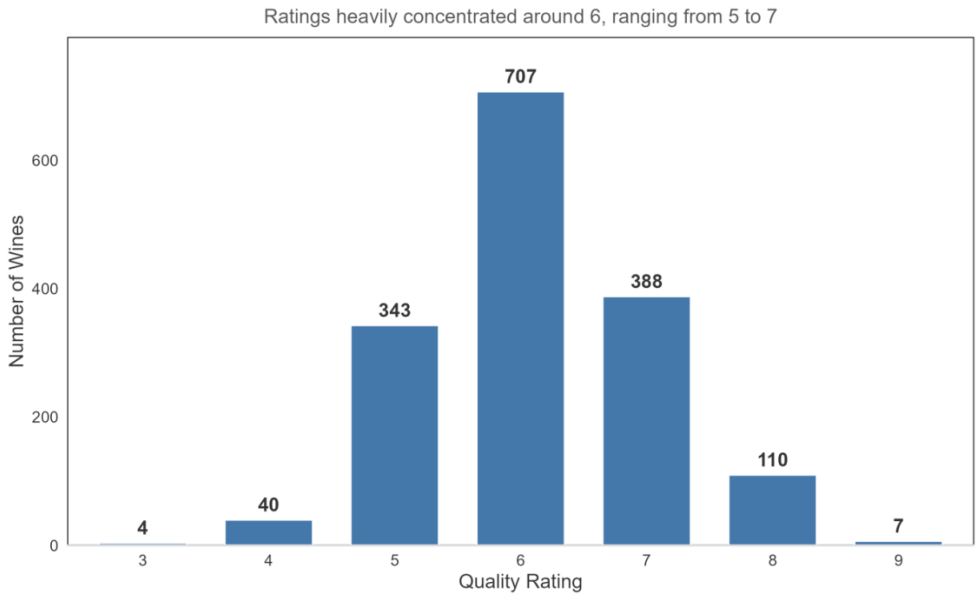

Figure 1: *Distribution of the Wine Quality Ratings*

Correlations with quality (figure 2) reveal alcohol as the strongest positive association (r = 0.451). Volatile acidity exhibits the strongest negative correlation (r = -0.361). Citric acid shows a moderate positive correlation, while density and fixed acidity exhibit high inter-correlation, potentially leading to multicollinearity in modelling. These initial patterns suggest physicochemical properties contain useful information about quality, but the moderate correlations indicate limits to purely objective prediction.

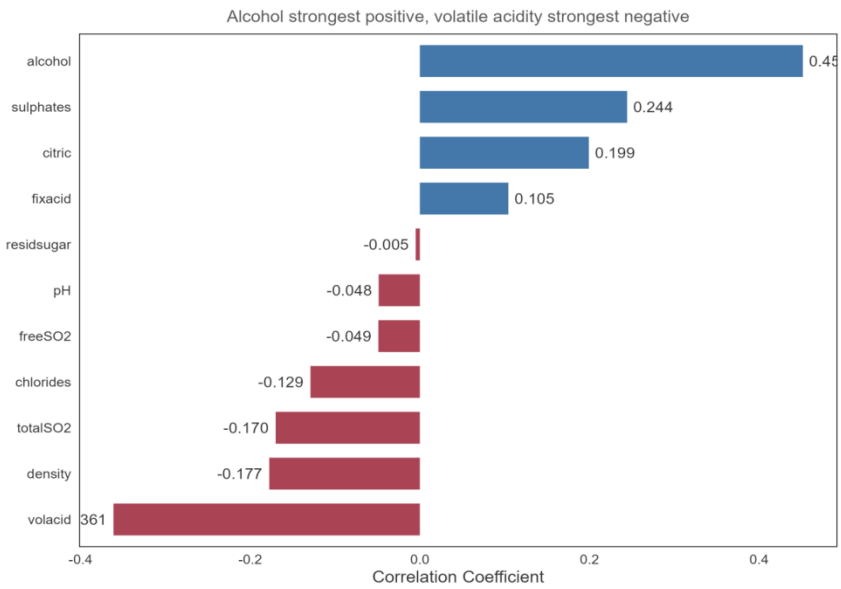

Figure2: *Correlations with the Wine Quality*

Scatter plots of the primary drivers (figure 3) confirm the trends. Higher alcohol content is associated with elevated quality ratings, while increased volatile acidity corresponds to lower ratings.

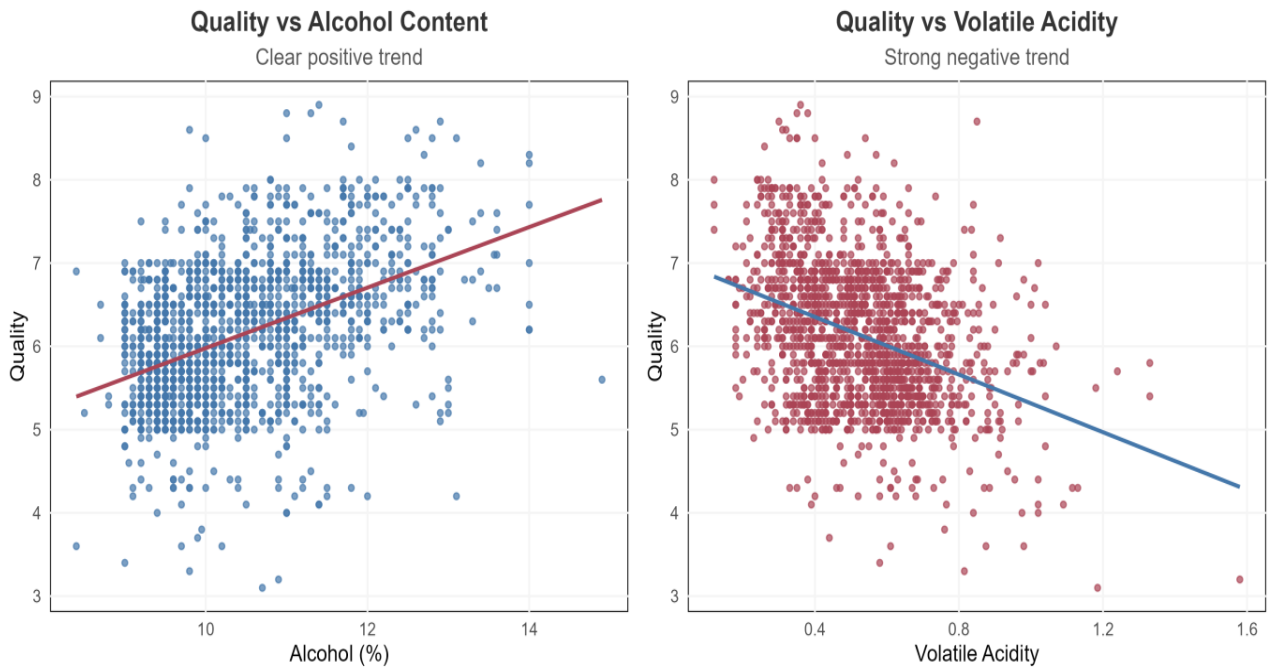

Figure 3: *Key Relationships Driving the Wine Quality*

1.2 Model Development and Selection

Multiple linear regression was employed to predict quality. The full model, using all 11 predictors, accounted for 32.2 % of the variance (R^2=0.322, adjusted R^2=0.317, residual standard error =0.709). Significant predictors ($p < 0.05$) included volatile acidity (-1.065), chlorides (-2.090), total sulphur dioxide (-0.003), pH (-0.479), sulphates (0.994), and alcohol (0.284).

Variance inflation factors highlighted moderate multicollinearity in the full model. Backward elimination, guided by statistical significance and VIF values, removed fixed acidity, residual sugar, and density, resulting in an eight predictor model (volatile acidity, citric acid, chlorides, free sulphur dioxide, total sulphur dioxide, pH, sulphates, alcohol) with equivalent performance (R^2 = 0.322, adjusted R^2 = 0.318, residual standard error = 0.708). All VIF values fell below 2, resolving multicollinearity concerns.

This eight predictor model represents the optimal balance. It is the simplest formulation that retains the full explanatory and predictive capability of the dataset. The retained predictors are also chemically interpretable, reinforcing confidence in the model's practical relevance.

1.3 Validity and Predictive Ability of the Model

Residual diagnostics (figure 4a and 4b) confirm the model assumptions are satisfied. The residuals versus fitted values and scale-location plots show random scatter and a flat trend, indicating homoscedasticity. While the Shapiro-Wilk test was significant (p= 0.001), visual inspection of the Q-Q plot and histogram showed only mild deviation from normality, which is unlikely to seriously affect the regression given the large sample size (n=1,599). The Breusch-Pagan test indicated some heteroscedasticity ($p < 0.001$), but residual plots showed no strong funnel shape, suggesting it is minor and the model remains practically useful. Not any transformation was needed as patterns are not severe.

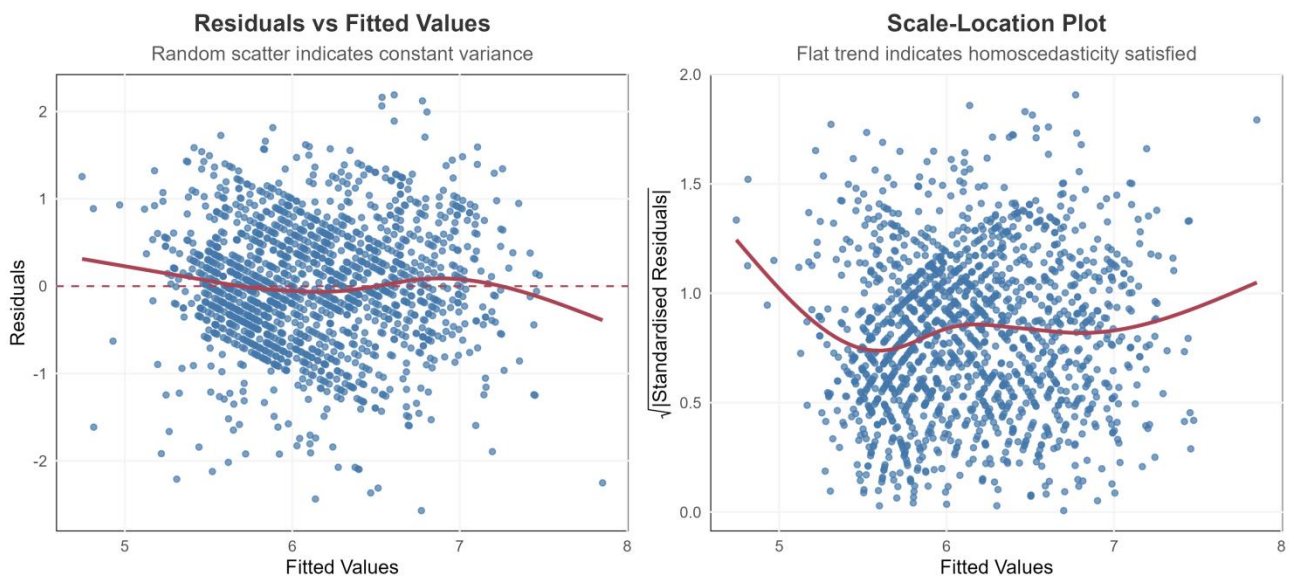

Figure 4a: *Checking Constant Variance (Homoscedasticity)*

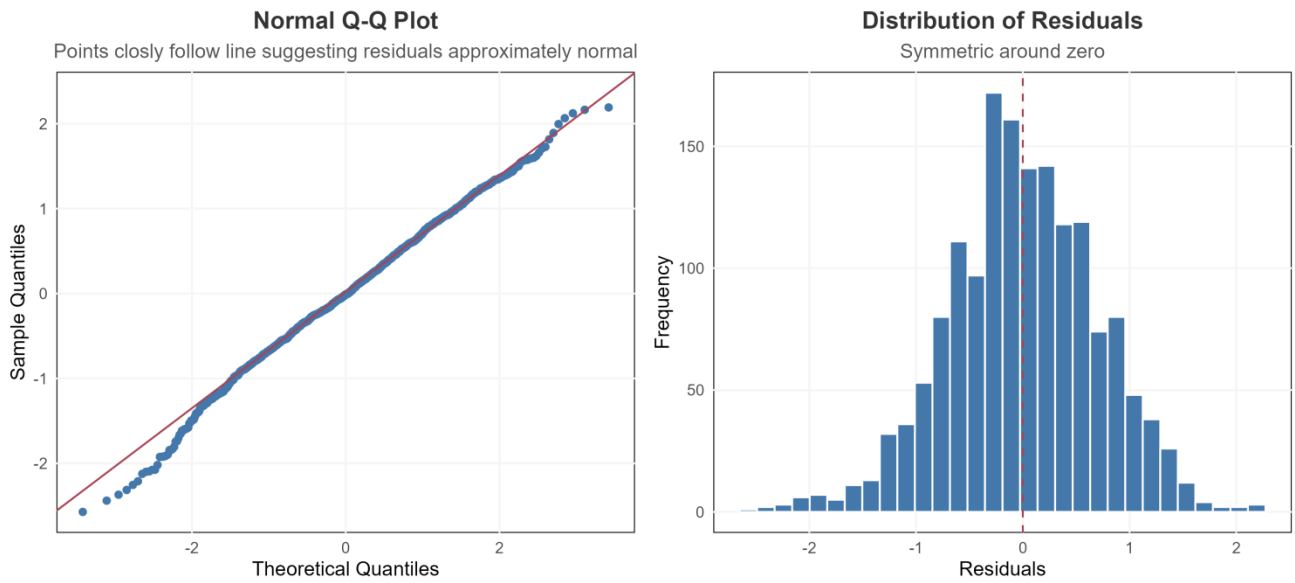

Figure 4b: *Checking Normality of Residuals*

The actual versus predicted plot (figure 5) illustrates reasonable calibration, with predictions closely following the line of perfect agreement, though with increased scatter at the extremes.

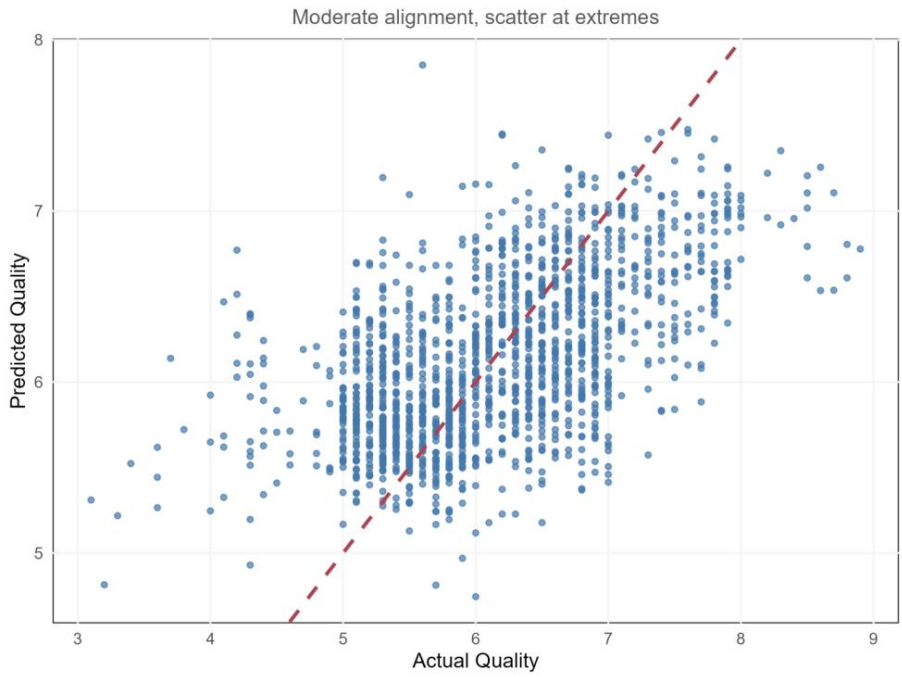

Figure 5: *Actual versus Predicted Quality*

The residual standard error of 0.708 translates to an average prediction error of approximately 0.71 quality points. An average error of approximately 0.71 points means the model is useful for ranking groups of wines (e.g., good versus average) but not for precise grading of individual bottles. While the model reliably captures key trends particularly the beneficial role of alcohol and the detrimental effect of volatile acidity, it explains only 32% of the variation. The unexplained portion likely reflects sensory attributes beyond laboratory measurement.

1.4 Conclusion

Physicochemical tests offer a moderate (explaining 32% of variance) but meaningful prediction of expert quality ratings. The eight predictor model provides an efficient validated tool suitable for automated preliminary assessment in wine production and regulatory contexts. Alcohol emerges as the primary positive driver, with low volatile acidity equally critical for higher ratings. Although not a complete substitute for human expertise, the model demonstrates practical utility for identifying influential factors and supporting data-driven decisions.

2. Predicting Hip Fracture from Bone Mineral Density and Patient Information

2.1 Exploration of the Association between Bone Mineral Density and Hip Fracture

The dataset includes 169 patients presenting with hip pain or injury, of whom 50 (29.6 %) had a fracture and 119 (70.4 %) did not (note the class imbalance). Bone mineral density (BMD) shows a marked difference: mean BMD is 0.850 in non-fracture cases and 0.623 in fracture cases.

Figure 6 illustrates this relationship clearly. The boxplot reveals substantial separation, with the median BMD for fracture cases around 0.58 and for non-fracture cases around 0.85, and minimal overlap between groups.

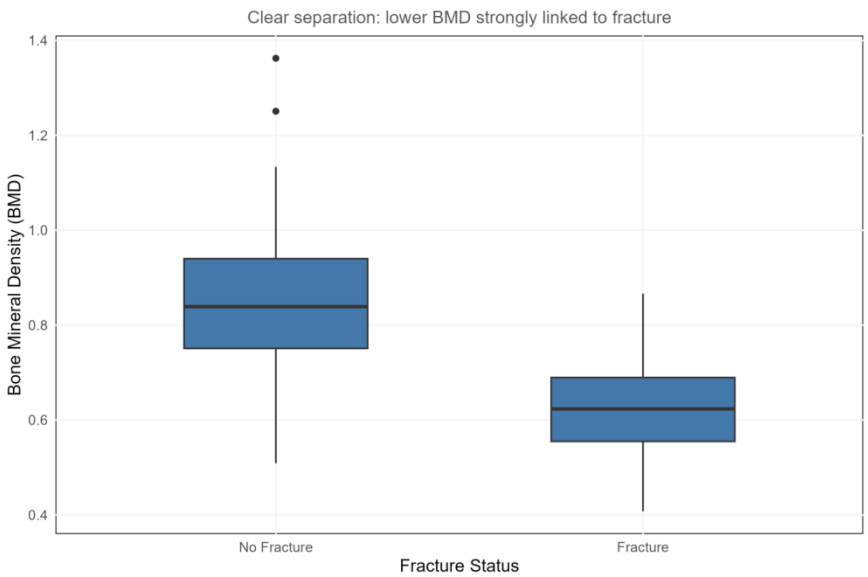

Figure 6: *Bone Mineral Density by Hip Fracture Status*

A simple logistic regression using BMD alone produces a highly significant negative coefficient (-15.785, $p < 0.001$) and an AUC of 0.909. For every 0.1 unit decrease in BMD, the odds of fracture increase by approximately a factor of 4.85. BMD alone provides strong predictive ability despite the class imbalance.

2.2 Model Development and Selection for Predicting Hip Fracture

Logistic regression was applied to evaluate the collective contribution of BMD and the other variables (age, sex, weight_kg, height_cm, medication, waiting_time). The full model retained BMD as the dominant predictor (-17.687, $p < 0.001$), with medication borderline significant (-1.295, $p = 0.053$).

Backward elimination removed waiting_time, weight_kg, sex, and age (non-significant after adjusting for BMD), resulting in a three predictor model: height_cm (+0.075, $p = 0.030$), medication (-1.291, $p = 0.050$), and BMD (-19.042, $p < 0.001$). This model has a lower AIC (114.69 versus 120.92 for the full model) and an AUC of 0.926.

Variables were removed in order of least significance, with each step improving AIC. Age was removed as its apparent effect is largely mediated through BMD. The final three predictor model is the best as it is efficient, statistically superior, and retains excellent predictive performance.

2.3 Validity and Predictive Ability of the Model

Variance inflation factors are low (all < 2), indicating no problematic multicollinearity. The model converges well, supporting validity. The AUC of 0.926 reflects very good discrimination. The improvement in AUC from 0.909 (BMD alone) to 0.926 confirms that height and medication add meaningful predictive value.
To illustrate practical application:

- A patient of height 165 cm, not on any medication, with BMD 0.9 has a predicted fracture probability of approximately 0.027 (very low risk).
- Whereas the same patient with BMD 0.7 has a probability of approximately 0.557 (high risk).

These examples show how the model can stratify patient risk in practice, guiding decisions such as preventive measures for high-risk cases.

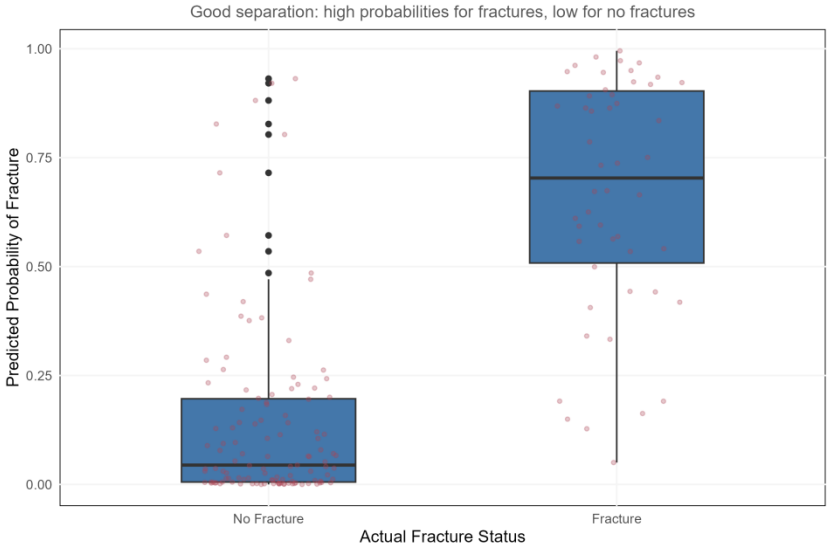

Figure 7: *Predicted Fracture Probability by Actual Status*

The plot (figure 7) shows good separation: predicted probabilities are high for actual fractures and low for non-fractures, supporting strong predictive ability.

The model is valid and offers strong predictive ability for clinical risk assessment, primarily driven by BMD, with height and medication providing useful supplementary information.

2.4 Conclusion

The analysis confirms BMD as the dominant predictor of hip fracture risk, with a simple three predictor model (BMD, height, medication) achieving very good discrimination (AUC 0.926). This model provides a practical, interpretable tool for identifying high-risk patients in clinical settings, supporting targeted preventive interventions.

References:

- Fox, J. et al. (2024) car: Companion to Applied Regression. Version 3.1-3. Available at: https://CRAN.R-project.org/package=car (Accessed: 30 December 2025).

- numiqo (2024) Regression Analysis | Full Course 2025. YouTube video, 146K views. Available at: https://www.youtube.com/watch?v=T5AoqxQFkzY (Accessed: 30 December 2025).

- STHDA (n.d.) QQ-plots: Quantile–Quantile plots – R Base Graphs. Available at: https://www.sthda.com/english/wiki/qq-plots-quantile-quantile-plots-r-base-graphs (Accessed: 22 December 2025).

Appendix I: Acknowledgement of GAI Contribution

This work is **entirely my own**. All key ideas, analysis, and reasoning were written by me. The **use of AI was carried out in line with the permitted guidance** in the Kingston University Student Guide to GAI for 2025/26.

Appendix II

R Codes

#1 Wine

```
library(tidyverse)
library(corrplot)
library(car)
wine <- read.csv("redwine.csv")

#fig 1
p_quality <- ggplot(wine_plot, aes(x = factor(quality_rounded))) +
  geom_bar(fill = "#4477aa", colour = "#ffffff", width = 0.7) +
  geom_text(stat = "count", aes(label = after_stat(count)),
            vjust = -0.6, size = 4.5, colour = "#333333", fontface = "bold") +
  labs(subtitle = "Ratings heavily concentrated around 6, ranging from 5 to 7",  # Updated
       x = "Quality Rating", y = "Number of Wines") +
  scale_y_continuous(expand = expansion(mult = c(0, 0.12))) +
  scale_x_discrete(labels = c("3", "4", "5", "6", "7", "8", "9")) +
  theme_minimal(base_size = 13) +
  theme(panel.grid = element_blank(),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5),
        axis.title = element_text(colour = "#333333"),
        axis.text = element_text(colour = "#333333"),
        axis.line.x = element_line(colour = "#dddddd"))

ggsave("1_quality_dist.png", p_quality, width = 9, height = 5.5, dpi = 300)

# fig2
quality_corrs <- sort(cor(wine)[,"quality"][-12])
cor_df <- data.frame(feature = names(quality_corrs), corr = quality_corrs)
p_cor <- ggplot(cor_df, aes(x = reorder(feature, corr), y = corr, fill = corr > 0)) +
  geom_col(width = 0.7) +
  coord_flip() +
  scale_fill_manual(values = c("TRUE" = "#4477aa", "FALSE" = "#aa4455"), guide = "none") +
  geom_text(aes(label = sprintf("%.3f", corr)),
            hjust = ifelse(cor_df$corr > 0, -0.15, 1.15), size = 4.2, colour = "#333333") +
  labs(subtitle = "Alcohol strongest positive, volatile acidity strongest negative",
       x = "", y = "Correlation Coefficient") +
  theme_minimal(base_size = 13) +
  theme(panel.grid = element_blank(),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5),
        axis.title = element_text(colour = "#333333"),
        axis.text = element_text(colour = "#333333"))
ggsave("2_correlation.png", p_cor, width = 9, height = 6.5, dpi = 300)

# fig3
p_alcohol <- ggplot(wine, aes(x = alcohol, y = quality)) +
```

```
  geom_point(alpha = 0.7, colour = "#4477aa", size = 1.4) +
  geom_smooth(method = "lm", colour = "#aa4455", se = FALSE, size = 1.1) +
  labs(title = "Quality vs Alcohol Content",
       subtitle = "Clear positive trend",
       x = "Alcohol (%)", y = "Quality") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

p_volacid <- ggplot(wine, aes(x = volacid, y = quality)) +
  geom_point(alpha = 0.7, colour = "#aa4455", size = 1.4) +
  geom_smooth(method = "lm", colour = "#4477aa", se = FALSE, size = 1.1) +
  labs(title = "Quality vs Volatile Acidity",
       subtitle = "Strong negative trend",
       x = "Volatile Acidity", y = "Quality") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

p_scatter <- p_alcohol + p_volacid +
  plot_annotation()
ggsave("fig3_scatter.png", p_scatter, width = 12, height = 5.5, dpi = 300)


full_model <- lm(quality ~ ., data = wine)
best_model <- step(full_model, direction = "backward", trace = FALSE)

pred <- predict(best_model)

# fig4
p_pred <- ggplot(data.frame(actual = wine$quality, predicted = pred), aes(x = actual, y = predicted))
+
  geom_point(alpha = 0.7, colour = "#4477aa", size = 1.4) +
  geom_abline(intercept = 0, slope = 1, colour = "#aa4455", linetype = "dashed", size = 1.1) +
  labs( subtitle = "Moderate alignment, scatter at extremes",
       x = "Actual Quality", y = "Predicted Quality") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))
ggsave("4_winePrediction.png", p_pred, width = 8, height = 6, dpi = 300)
```

```
res_data <- data.frame(fitted = fitted(best_model), residuals = resid(best_model), std_res = rstandard(best_model))

#5
p_res_fitted <- ggplot(res_data, aes(x = fitted, y = residuals)) +
  geom_point(colour = "#4477aa", alpha = 0.7, size = 1.5) +
  geom_hline(yintercept = 0, colour = "#aa4455", linetype = "dashed") +
  geom_smooth(se = FALSE, colour = "#aa4455") +
  labs(title = "Residuals vs Fitted Values",
       subtitle = "Random scatter indicates constant variance holds",
       x = "Fitted Values", y = "Residuals") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

p_scale <- ggplot(res_data, aes(x = fitted, y = sqrt(abs(std_res)))) +
  geom_point(colour = "#4477aa", alpha = 0.7, size = 1.5) +
  geom_smooth(se = FALSE, colour = "#aa4455") +
  labs(title = "Scale-Location Plot",
       subtitle = "Flat trend indicates homoscedasticity satisfied",
       x = "Fitted Values", y = expression(sqrt("|Standardised Residuals|"))) +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

p_homo <- p_res_fitted + p_scale
ggsave("5a_homosced.png", p_homo, width = 12, height = 5.5, dpi = 300)

# b
p_qq <- ggplot(res_data, aes(sample = residuals)) +
  stat_qq(colour = "#4477aa", size = 1.8) +
  stat_qq_line(colour = "#aa4455") +
  labs(title = "Normal Q-Q Plot",
       subtitle = "Points follow line suggest residuals approximately normal",
       x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

p_hist <- ggplot(res_data, aes(x = residuals)) +
  geom_histogram(bins = 30, fill = "#4477aa", colour = "white") +
  geom_vline(xintercept = 0, colour = "#aa4455", linetype = "dashed") +
  labs(title = "Distribution of Residuals",
```

```
      subtitle = "Symmetric around zero",
      x = "Residuals", y = "Frequency") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

p_normal <- p_qq + p_hist
ggsave("5b_normali.png", p_normal, width = 12, height = 5.5, dpi = 300)


# 2 BMD
library(tidyverse)
library(pROC)
bmd <- read.csv("bmd-1.csv")
table(bmd$fracture) / nrow(bmd)

p_bmd_box <- ggplot(bmd, aes(x = factor(fracture, labels = c("No Fracture", "Fracture")), y = bmd))
+
  geom_boxplot(fill = "#4477aa", colour = "#333333", width = 0.5) +
  labs( subtitle = "Clear separation: lower BMD strongly linked to fracture",
      x = "Fracture Status", y = "Bone Mineral Density (BMD)") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_line(colour = "#f5f5f5"),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "white"),
        plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(colour = "#555555", hjust = 0.5))
ggsave("bmd_boxplot.png", p_bmd_box, width = 9, height = 6, dpi = 300)

bmd_only <- glm(fracture ~ bmd, data = bmd, family = binomial)
summary(bmd_only)
roc_bmd <- roc(bmd$fracture, predict(bmd_only, type = "response"))
auc(roc_bmd)

full_logit <- glm(fracture ~ age + sex + weight_kg + height_cm + medication + waiting_time + bmd,
                  data = bmd, family = binomial)
summary(full_logit)

best_logit <- step(full_logit, direction = "backward", trace = FALSE)
summary(best_logit)
auc(roc(bmd$fracture, predict(best_logit, type = "response")))

# for the illustration part
new_low_risk <- data.frame(height_cm = 165, medication = 0, bmd = 0.9)
predict(best_logit, new_low_risk, type = "response")

new_high_risk <- data.frame(height_cm = 165, medication = 0, bmd = 0.7)
predict(best_logit, new_high_risk, type = "response")
```

```
pred_prob <- predict(best_logit, type = "response")

pred_data <- data.frame(actual = factor(bmd$fracture, labels = c("No Fracture", "Fracture")),
  predicted_prob = pred_prob)

p_pred_prob <- ggplot(pred_data, aes(x = actual, y = predicted_prob)) +
  geom_boxplot(fill = "#4477aa", colour = "#333333", width = 0.5) +
  geom_jitter(alpha = 0.3, colour = "#aa4455", width = 0.2, size = 1.2) +
  labs( subtitle = "Good separation: high probabilities for fractures, low for no fractures",
      x = "Actual Fracture Status", y = "Predicted Probability of Fracture") +
  theme_minimal(base_size = 13) +
  theme(panel.grid.minor = element_blank(),
      panel.grid.major = element_line(colour = "#f5f5f5"),
      plot.background = element_rect(fill = "white"),
      panel.background = element_rect(fill = "white"),
      plot.title = element_text(colour = "#333333", hjust = 0.5, face = "bold"),
      plot.subtitle = element_text(colour = "#555555", hjust = 0.5))

ggsave("bmd_prediction.png", p_pred_prob, width = 9, height = 6, dpi = 300)
```